REDBIRD FLIGHT SIMULATIONS
QUALIFICATION AND APPROVAL GUIDE (QAG)

MDN AND MRG AMENDED QUALIFICATION AND APPROVAL GUIDE, VERSION 1.1



Advanced Aviation Training Device

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FAA APPROVED QAG

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SECTION 1: OVERVIEW STATEMENT

This Qualification and Approval Guide (QAG) includes a detailed description of all components, functions, and capabilities for the Redbird model MDN/MRG. This includes any optional aircraft configurations with quality color pictures and diagrams. The required information as described in the current advisory circular, AC 61-136A / FAA Approval of Aviation Training Devices (ATD) and Their Use for Training and Experience, required for FAA approval, is contained within this document. This includes listing all the required qualifying items, functions, and capabilities. This QAG is used to clearly describe and verify the functionality of this aviation training device platform confirming its suitability for airman training and experience. A valid FAA Letter of Authorization (LOA) specifying the ATD allowances must also accompany the training device when utilized for airman training or experience requirements as specified in 14 CFR §61 or 141.

This ATD must continue to maintain its performance and function without degradation. The operator of this device is expected to maintain its condition and functionally when used for airmen training or experience requirements. Only the configurations approved for this model, as described within this QAG, can be utilized when satisfying FAA experience or training requirements. Any additions, changes, or revisions to the model or to the available configurations described within this document must be approved in writing by The General Aviation and Commercial Division, AFS-800.

Any questions concerning FAA approval of ATDs should be directed to the General Aviation and Commercial Division, AFS-800.

SECTION 2: OVERALL AVIATION TRAINING DEVICE (ATD) DESCRIPTION AND PICTURES

The Redbird model MDN/MRG is based on the dimensions and layout of a Piper Malibu production single engine-land, turboprop and piston aircraft variants. The MDN/MRG closely represents the overall functionality, performance, avionics, and instrumentation. The platform consists of a cockpit section, instructor's control station, visual display system and an audio system. It incorporates a combination of hardware and software components that is assembled and checked by Redbird Flight Simulations. All hardware elements are permanently installed and designed so the cockpit has the appearance and feel of an actual aircraft. From the pilot's seated position, there are no computer hardware elements such as keyboards, pointing devices, etc. for his or her use.

The MDN/MRG provides realistic and true-to-scale cockpit design, avionics, and reliable hardware/software performance. This platform also provides an effective training environment for student and certificated pilots. This includes the capability of practicing scenario based flight training events, simulated equipment failure and emergency procedures, pilot evaluations, instrument procedures/ experience, and facilitating increased pilot proficiency overall.

Single Engine-Land, Turbo Prop/Piston Land with Enclosed Cockpit



Image 1: General view of Motion Platform and Cockpit Enclosure

The Redbird MDN/MRG is a versatile and affordable device that was designed to represent a variety of Piper PA46 Malibu Models. It is equipped with the following notable features:

- 3-axis electric motion platform providing pitch, roll and yaw motions.
- Wrap-around exterior visuals provided by 6 LCD screens
- Dual pilot controls including a 2-axis control-loaded yoke and interconnected rudder pedals
- Realistic switches, buttons, knobs, circuit breakers and other cockpit controls that are designed to match the aircraft wherever possible
- An interchangeable instrument panel to allow development of future configurations
- Closed Circuit intercom system, allowing for communication between the pilot, co-pilot and instructors using standard aviation headsets
- A portable instructors station, allowing the instructor to operate from inside or outside the simulator
- Optional supplemental oxygen system for the pilot and/or co-pilot

General Photographs



Image 2: General view of MDN/MRG-E1 (Avidyne Instrument Panel)



Image 3: General view of MDN/MRG-G1 (Garmin G1000 Instrument Panel)



Image 4: General view of Pilot Yoke



Image 5: Detailed view of Pilot Yoke Switches (Left Side)



Image 6: General view of Co-Pilot Yoke



Image 7: Detailed view Co-Pilot Yoke Switches (Right Side)

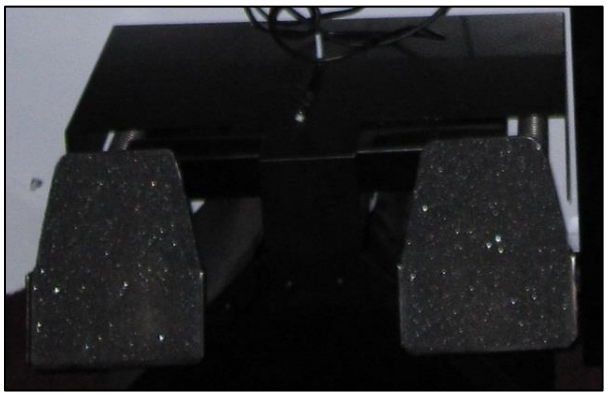


Image 8: Detailed view of Pilot/Co-Pilot Rudder Pedals



Image 9: General view of Instructors Station Tablet PC

MDN (Meridian) Specific Photographs



Image 10: Motion Platform and Cockpit Enclosure (Front)



Image 11: Motion Platform and Cockpit Enclosure (Rear)



Image 12: General view of Pilot Lower Switch Panel (Left Side)



Image 13: General view of Pilot Circuit Breaker Panel

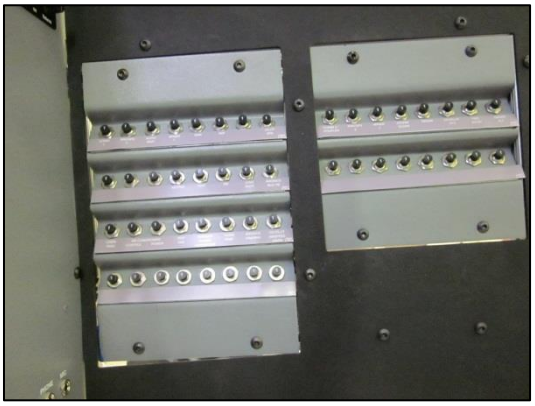


Image 14: General view of Co-Pilot Circuit Breaker Panel



Image 15: General view of FMS Console and Keypad

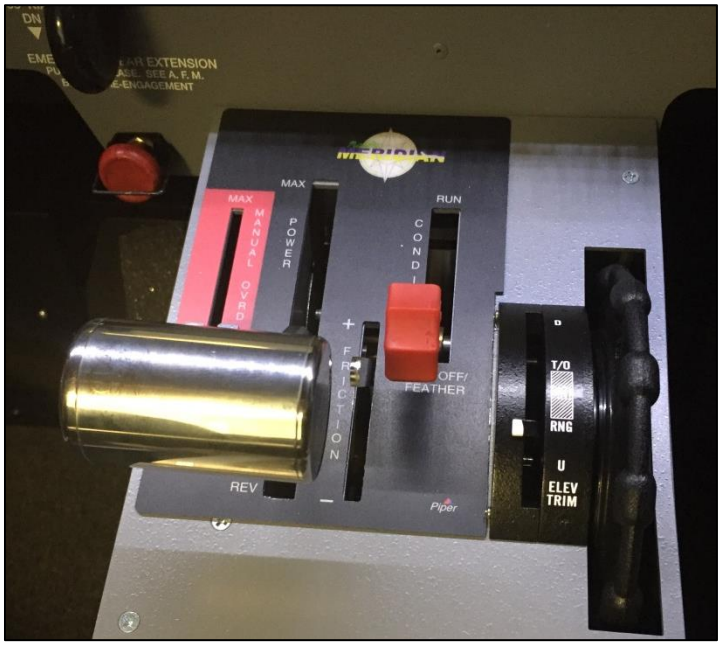


Image 16: Detailed view of Throttle Quadrant



Image 17: General interior view of assembled cockpit (Center View)

MRG (Matrix/Mirage) Specific Photographs



Image 18: Motion Platform and Cockpit Enclosure (Front)



Image 19: Motion Platform and Cockpit Enclosure (Rear)



Image 20: General view of Pilot Yoke Switches



Image 21: Detailed view of Pilot Lower Switch Panel (Left)



Image 22: Detailed view of Pilot Lower Switch Panel (Right)



Image 23: Co-Pilot Lower Switch Panel (Right)



Image 24: General view of Pilot Circuit Breaker Panel



Image 25: General view of Co-Pilot Circuit Breaker Panel



Image 26: General view of Overhead Switch Panel



Image 27: Detailed view of Pilot Overhead Switches



Image 28: Detailed view of Co-Pilot Overhead Switches



Image 29: Detailed view of Air Induction Panel



Image 30: General view of FMS Console and Keypad

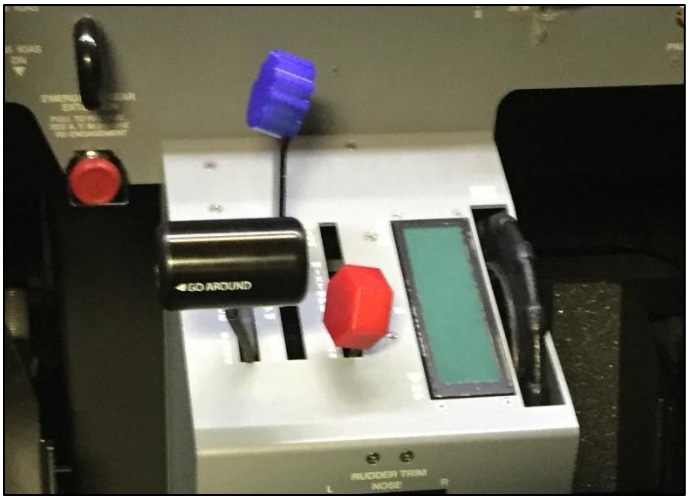


Image 31: Detailed view of Throttle Quadrant



Image 32: General interior view of assembled cockpit (Center View)

SECTION 3: TRAINING DEVICE COMPONENT LIST

MDN/MRG Component Listing

Item	Component Name	Manufacturer	Type	Qty	Description/Function	Configuration
1	Motion Platform	Redbird Flight Simulations	Hardware	1	Gimbaled, steel motion platform with movement in pitch, roll and yaw. Includes all motors, sensors and safety controls.	All
2	Cockpit Enclosure	Redbird Flight Simulations	Hardware	1	Cockpit enclosure to exclude distractions.	All
3	Dual Yoke	Redbird Flight Simulations	Hardware/ Software	1	Dual, Control loaded pitch and roll controller with switches and buttons for airplane systems operation.	All
4	Dual Rudder Pedals	Redbird Flight Simulations	Hardware	1	Dual Rudder control pedals with toe brakes.	All
5	Throttle Quadrant	Redbird Flight Simulations	Hardware/ Software	1	Single, turbine engine controls with throttle lever, condition lever, manual override lever, go around, bleed air, trim wheel and rudder trim controls.	MDN
6	Throttle Quadrant	Redbird Flight Simulations	Hardware/ Software	1	Single, piston engine controls with throttle lever, mixture lever, propeller pitch lever, go around, bleed air, trim wheel and rudder trim controls.	MRG
7	Instrument Controls	Redbird Flight Simulations	Hardware/ Software	1	Flight instruments, radios, airplane configuration and systems controls. Versions currently exist for Avidyne & G1000	All
8	Simulation Computer	Redbird Flight Simulations	Hardware	1	Host computer for flight simulation engine, simulation control software, airplane systems and instruments.	All
9	Motion Control Computer	Redbird Flight Simulations	Hardware	1	Host computer for FMX Control.	All
10	Switch Panel 1	Redbird Flight Simulations	Hardware/ Software	1	Lower switch panel with airplane configuration and systems controls.	All
11	Switch Panel 2	Redbird Flight Simulations	Hardware/ Software	1	Overhead switch panel with airplane systems controls.	All
12	Circuit Breaker Panel	Redbird Flight Simulations	Hardware/ Software	1	Circuit breaker controls for pilot side, copilot side and front panel.	All

Item	Component Name	Manufacturer	Type	Qty	Description/Function	Configuration
13	FMS Panel	Redbird Flight Simulations	Hardware	1	FMS controls, keypad and display screen. Includes airplane cabin pressurization system controls.	All
14	Redbird Avidyne	Mindstar Aviation/Redbird	Software	1	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	All
15	Redbird G1000	Mindstar Aviation/Redbird	Software	1	Virtual flight instruments, radios, gauges, indicators, alerts, misc. instruments and logic controls for simulated systems.	All
16	LCD	22” Industry Standard	Hardware	6	Flat Panel displays for exterior views.	All
17	LCD	19” Industry Standard	Hardware	3	Flat Panel displays for virtual instruments.	All
18	FMX Control	Redbird Flight Simulations	Software	1	Motion system control and component integration. (depending on serial number)	All
19	Windows 7	Microsoft	Software	1	Operating system.	All
20	ESP	Microsoft	Software	1	Simulation engine. (depending on serial number)	All
21	Prepar3D	Lockheed Martin	Software	1	Simulation engine. (depending on serial number)	All
22	RB Sim	Redbird Flight Simulations	Software	1	Simulation control and component integration.	All
23	Instrument Procedures Database	RealNav Data	Software	1	Provides for FAA published instrument navigation procedures, database per 14 CFR 97 (enroute, approach)	All
24	Instructors Station Computer	Industry Standard	Hardware	1	Host computer for Instructors Station.	All
25	Instructors Station	Redbird Flight Simulations	Software	1	Environmental, location and failure controls with map, track and glideslope display.	Optional
26	Navigator	Redbird Flight Simulations	Software	1	Environmental, location and failure controls with map, track and glideslope display.	Optional
27	Parrot	Redbird Flight Simulations	Software	1	Optional software package that links iOS aviation or navigational apps to Redbird Devices	Optional
28	Cygnus	Redbird Flight Simulations	Software	1	Optional software package providing ATC Services	Optional

Item	Component Name	Manufacturer	Type	Qty	Description/Function	Configuration
29	Insight	Redbird Flight Simulations	Software	1	Optional package that provides instructors live video feed of the cockpit and instrumentation through Prepar3D software	Optional

Table 1: Training Device Component List (Shown Above)

Statements of Compatibility of Software and Hardware AC61-136A, Appendix 2, 4. b.:

The device and all configurations included in this Qualification and Approval Guide (QAG) meet the requirements for the Compatibility of Software and Hardware.

Microsoft Windows 7 (Operating System)

"This is to certify that Microsoft Corporation, the owner and developer of the Windows 7 operating system, has evaluated that their operating system works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc., utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software."

Lockheed Martin Prepar3D (Simulation Engine)

This is to certify that Lockheed Martin, the owner and developer of the Prepar3D Simulator Engine, has evaluated that their software application works with industry standard PC's and USB flight control devices. Redbird Flight Simulations, Inc. utilizes industry standard USB flight control devices for all pilot input. All input control devices meet the USB 2.0 industry standard specified interfaces. Redbird Flight Simulations, Inc., the component integrator, has determined that the transport delay time is less than 300 milliseconds, and that all analog and digital input signals meet the performance criteria established for the software.

SECTION 4: AVIATION TRAINING DEVICE (ATD) DESIGN CRITERIA LIST

The following section provides the detailed “word for word” listing and design criteria of each of the required items, functions, and capabilities (listed in AC 61-136A, for BATD requirements Appendix 2 and the additional AATD items of Appendix 3) and operational performance value/scale (as applicable) for each of the functions described for the Redbird MDN/MRG.

DESCRIPTION OF COMPLIANCE ITEMS AND FUNCTIONS

BATD DESIGN CRITERIA

All configurations for this model, as noted, meet AC 61-136A, Appendix 2 requirements

a. Redbird MDN/MRG model meets the following Control Input requirements.

(1) Physical flight and aircraft control systems are designed such that they are recognizable as to their function and how they are manipulated solely from their appearance. No keyboard, mouse, or gaming joystick is used to control the aircraft.

(2) Virtual controls are limited to setting the aircraft configuration, location, weather conditions, and pausing action.

(3) No keyboard or mouse is used to set or position any of the below listed features of this device. The required and additional equipment is operated in the same manner in which it would be operated in the aircraft represented. **This includes power levers, propeller levers, mixture (MRG) or condition levers (MDN), landing gear, and wing flaps.**

(4) The physical arrangement, appearance, and operation of controls, instruments, and switches closely represents the model of aircraft including: **master switch, ignition switch (MDN) and start switches, battery, generators, EMER BATT switch, external power switch, avionics master, beacon/strobe lights, navigation lights, taxi and landing lights.** Also included are: **pitot heat, anti-ice switches, bleed air, fire detection test (MDN), overspeed governor test switch (MDN), standby fuel pumps.**

(5) Only the software package evaluated and approved by the FAA is available for use on this computer system.

b. “Redbird MDN/MRG” model meets the following Control Requirements:

(1) Airplane

(a) **Self-centering control yoke or control stick** that allows continuous adjustment of pitch and bank

(b) **Self-centering rudder pedals** rudder pedals that allow continuous adjustment of yaw and corresponding reaction in heading and roll.

(c) **Power Levers** that allows for continuous movement from idle to full power, with corresponding changes in pitch and yaw.

(d) **Mixture/Condition, propeller, and throttle/power control(s)** as applicable to the model aircraft represented

(e) **Controls** for the following items applicable to the aircraft represented:

- Wing flaps
- Pitch trim
- Communication and navigation radios
- Clock or timer
- Gear handle (if applicable)
- Transponder
- Altimeter

c. Redbird MDN/MRG model meets the following Control Input requirements:

(1) Time from control input to recognizable system response must be without delay (i.e., not appear to lag in any way). Redbird Flight Simulations, Incorporated verifies that the MDN/MRG meets this requirement.

(2) The control inputs are tested by the computer and software at each session startup and displayed as a confirmation message or a “warning message” that the transport delay time or any design parameter is out of original tolerances. It is not possible to continue the training session unless the problem is resolved and all components are functioning properly. This test considers the items listed in the display requirements.

d. Redbird MDN/MRG model meets the following Display Requirements:

(1) **Instruments and indicators** replicated and properly located as appropriate to the aircraft model represented:

Flight instruments in a standard configuration, represented as traditional “round dial” flight instruments, or as an electronic primary flight instrument display (PFD) and multi-function display (MFD) with reversionary and back-up flight instruments.

- A sensitive **altimeter** with incremental markings each 20 feet or less, operable throughout the normal operating range of the aircraft or family of aircraft represented.
- A **magnetic direction indicator**
- A **heading indicator** with incremental markings each 5 degrees or less, displayed on a 360 degree circle. Arc segments of less than 360 degrees may be selectively displayed if desired or required, as applicable to the aircraft or family of aircraft represented.
- An **airspeed indicator** with incremental markings as shown on the aircraft or family of aircraft represented; airspeed markings of less than 40 knots need not be displayed.
- A **vertical speed indicato**r with incremental markings each 100 feet per minute (fpm) for both climb and descent, for the first 1,000 fpm of climb and descent, and at each 500 fpm climb and descent for the remainder of a minimum ± 2,000 fpm total display, or as applicable to the aircraft or family of aircraft being represented.
- A **gyroscopic rate-of-turn indicator or equivalent** with appropriate markings for a rate of 3 degrees per second turn for left and right turns. If a turn and bank indicator is used, the 3 degrees per second rate index must be inside of the maximum deflection of the indicator.
- A **slip and skid indicator** with coordination information displayed in the conventional skid ball format where a coordinated flight condition is indicated with the ball in the center position. **A split image triangle indication** may be used if applicable to the aircraft or family of aircraft being represented.
- An **attitude indicator** with incremental markings each 5 degrees of pitch or less, from 20 degree pitch up to 40 degree pitch down or as applicable to the aircraft or family of aircraft represented. Bank angles must be identified at “wings level” and at 10, 20, 30, and 60 degrees of bank (with an optional additional identification at 45 degrees) in left and right banks.
- **Engine instruments** as applicable to the aircraft or family of aircraft being represented, providing markings for normal ranges and minimum and maximum limits.
- A suction gauge or **instrument pressure indicator**, as applicable, with a **display** applicable to the aircraft or family of aircraft represented.
- A **flap setting indicator** that displays the current flap setting. Setting indications should be typical of that found in an actual aircraft.
- A **pitch trim indicator** with a display that shows zero trim and appropriate indices of airplane nose down and airplane nose up trim, as would be found in an aircraft.
- **Communication radio(s)** with display(s) of the radio frequency in use.
- A **clock** with incremental markings for each minute and second, or a timer with a display of minutes and seconds,
- A **transponder** that displays the current transponder setting.
- A **fuel quantity indicator(s)** that displays the fuel remaining, either in analog or digital format,

as appropriate for the aircraft or family of aircraft represented.

Navigation Equipment: Replicates both precision and non-precision instrument navigation, including approach procedures (with an aural identification feature if normally available), and a marker beacon receiver. Examples: an ILS receiver, non-directional radio beacon (NDB), Global Positioning System (GPS), Localizer (LOC) or Very high frequency Omnidirectional Range (VOR). Graduated markings as indicated below are present on each course deviation indicator as applicable.

Markings include:

- One-half dot or less for **course/glide slope deviation indicator** (i.e., VOR, LOC, or ILS),
- 5 degrees or less for bearing deviation for automatic direction finder (ADF) and radio magnetic indicator (RMI), as applicable,

(2) All instrument displays listed above must be visible during all flight operations. The update rate of all displays must provide an image of the instrument that:

(a) Does not appear to be out of focus or illegible
(b) Does not appear to "jump" or "step" to a distracting degree during operation.
(c) Does not appear with distracting jagged lines or edges.
(d) Does not appear to lag relative to the action and use of the flight controls.

(3) Control inputs are reflected by the flight instruments in real time and without a perceived delay in action. Display updates or actions show all changes (within the total range of the replicated instrument) that are equal to or greater than the following:

(a) Airspeed indicator: change of 5 knots.
(b) Attitude indicator: change of 2 degrees in pitch and bank.
(c) Altimeter: change of 10 feet.
(d) Turn and bank: change of ¼ standard rate turn.
(e) Heading indicator: change of 2 degrees.
(f) Vertical speed indicator (VSI): change of 100 fpm.
(g) Tachometer: change of 25 rpm or 2 percent of turbine speed.
(h) VOR/ILS: change of 1 degree for VOR or ¼ of 1 degree for ILS.
(i) ADF: change of 2 degrees.
(j) GPS: change as appropriate for the model of GPS based navigator represented.
(k) Clock or timer: change of 1 second.

(4) **Flight Displays reflect proper dynamic behavior**. Examples: VSI reading of 500 fpm reflects a corresponding movement in altimeter. Throttle movement reflects an appropriate change in pitch, rpm indication, or the power indicator. Pitch changes reflect an appropriate change in airspeed and altitude.

e. Redbird MDN/MRG model meets the following Flight Dynamics requirements:

(1) All flight dynamics are comparable to the way the represented airplane performs and handles.

(2) All simulated aircraft performance parameters (such as maximum speed, cruise speed, stall speed, maximum climb rate and hovering/sideward/forward/rearward flight) are comparable to the airplane represented. A performance table is included in the QAG for each aircraft configuration for sea level and 5,000 feet.

(3) Vertical lift component changes as a function of bank/load factor, comparable to the aircraft represented.

(4) Changes in flap setting and gear position are accompanied by changes in flight dynamics comparable and reasonably equivalent to the aircraft represented.

(5) The presence and intensity of wind and turbulence must be reflected in the handling and performance qualities of the simulated aircraft and should be comparable to the way the aircraft or family of aircraft represented performs and handles.

f. Redbird MDN/MRG model meets the following Instructional Management Requirements:

(1) The instructor can pause the system at any point during the training session for the purpose of administering instruction regarding the training task at hand.

(2) If a training session begins with the “aircraft in the air” and ready for the performance of a particular procedural task, the instructor can manipulate the following system parameters independently of the simulation:

- Aircraft geographic location,
- Aircraft heading,
- Aircraft airspeed,
- Aircraft altitude, and
- Wind direction, speed, and turbulence.

(3) The **system is capable of recording** both a horizontal and vertical track of aircraft movement for the entire training session for later playback and review.

(4) The instructor can disable any of the instruments prior to or during a training session and simulate failure of any of the instruments without stopping the flight simulation to affect the failure.

(5) The ATD has an available **REAL NAV Approach Database** that is local to the training facility. All navigational data must be based on **procedures as published per 14 CFR Part 97.**

AATD DESIGN CRITERIA

All configurations, as noted in AC 61-136A, Appendix 3 meet these *additional* AATD design criteria items listed.

a. Redbird MDN/MRG model meets the following additional AATD CRITERIA.

(1) A realistic shrouded (enclosed) or unshrouded (open) cockpit design and instrument panel arrangement representing a specific model aircraft cockpit.

(2) **Cockpit knobs, system controls, switches, and/or switch panels in realistic sizes and design** appropriate to each intended functions, in the proper position and distance from the pilot's seated position, and representative of the category and class of aircraft being represented.

(3) Primary flight and navigation instruments appropriately sized and properly arranged that exhibit neither stepping nor excessive transport delay.

(4) Digital Avionics Panel

(5) IFR Moving Map **GPS Navigation System**

(6) **Two-axis autopilot**, and, as appropriate, **a flight director (FD)**. (Airplane only requirement)

(7) **Pitch trim** (manual or electric pitch trim) permitting indicator movement either electrically or analog in an acceptable trim ratio.

(8) **Visual system** providing cues for both day and night, including variable VFR and IFR conditions including:

- Adjustable visibility parameters; and
- Adjustable ceiling parameters.

(9) **Cockpit seat** is representative of the aircraft make/model.

(10) A **separate instructor station** that permits effective interaction without interrupting the flight in overseeing the pilot's horizontal and vertical flight profiles in real time and space. This includes the ability to:

(a) Oversee tracks along published airways, holding entries and patterns, and Localizer (LOC) and glideslope (GS) alignment/deviation (or other approaches with a horizontal and vertical track).
(b) Function as air traffic control in providing vectors, etc., change in weather conditions, ceilings, visibilities, wind speed and direction, light/moderate/severe turbulence, and icing conditions.

(c) Invoke failures in navigation and instruments, radio receivers, landing gear and flaps, engine power (partial and total), and other aircraft systems (pitot, electric, static, etc.) by using either a keyboard or mouse.

(12) Push to talk switch on the control yoke or stick

(13) Capable of simulating all of the emergency procedures for which a checklist is prescribed in the aircraft manufacturers' pilot operating handbook (POH) or flight manual

SECTION 5: AIRCRAFT CONFIGURATIONS

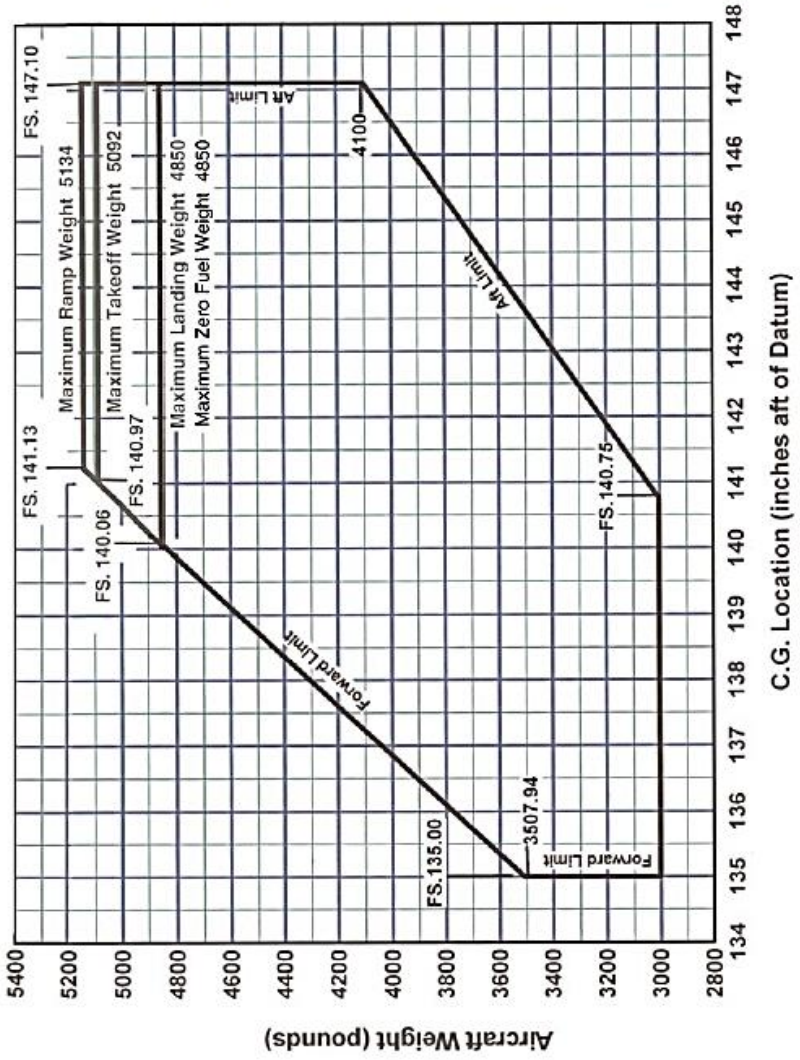
LIST OF PREVIOUSLY APPROVED CONFIGURATION(S):

1. **Configuration: MDN (PA46-500-E1)**
   a. **Cockpit**



   b. **Weight & Balance Data**

6.9 WEIGHT AND BALANCE DETERMINATION FOR FLIGHT (cont)



Center of Gravity Limits Graph
Figure 6-33

c. **Performance Data**

Altitude	MTOW	Max Operating Maneuvering Speed (V_O)	Never Exceed Speed (V_{NE})	Max Operating Speed (V_{MO})	Max Cruise Speed (V_C)	Stall Speed (V_S)	Max Power Rate of Climb
Sea Level	5,092 lbs.	127 KIAS	N/A	188 KIAS	183 KTAS at 969 ft-lb	69 KIAS with Flaps-Down	1610 FPM at 125 KIAS
5,000 feet	5,092 lbs.	127 KIAS	N/A	188 KIAS	197 KTAS at 1030 ft-lb	69 KIAS with Flaps-Down	1460 FPM at 125 KIAS

Table 2: MDN Avidyne Configuration Performance Data
(Note: Data represents Standard Atmospheric Conditions at maximum gross weight)

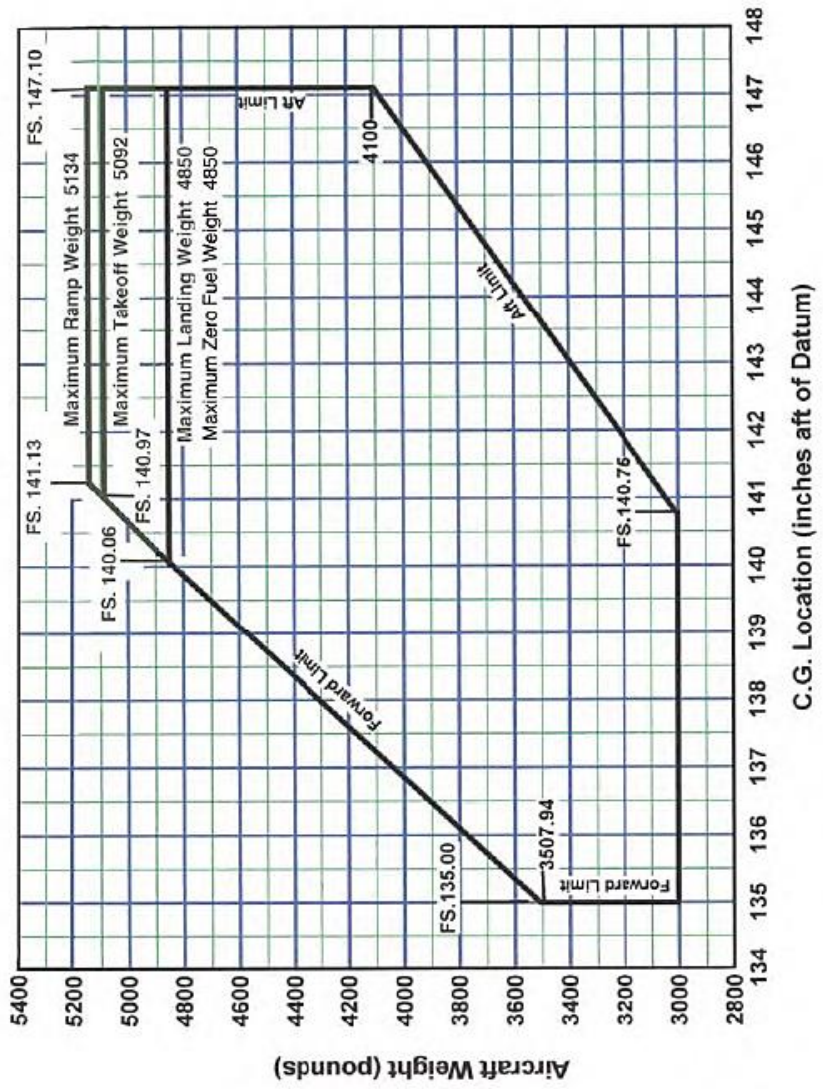
2. Configuration: MDN (PA46-500-G1)

a. **Cockpit**



b. **Weight & Balance Data**

6.9 WEIGHT AND BALANCE DETERMINATION FOR FLIGHT (cont)



Center of Gravity Limits Graph
Figure 6-33

c. **Performance Data**

Altitude	MTOW	Max Operating Maneuvering Speed (V_O)	Never Exceed Speed (V_{NE})	Max Operating Speed (V_{MO})	Max Cruise Speed (V_C)	Stall Speed (V_S)	Max Power Rate of Climb
Sea Level	5,092 lbs.	127 KIAS	N/A	188 KIAS	183 KTAS at 969 ft-lb	69 KIAS with Flaps-Down	1610 FPM at 125 KIAS
5,000 feet	5,092 lbs.	127 KIAS	N/A	188 KIAS	197 KTAS at 1030 ft-lb	69 KIAS with Flaps-Down	1460 FPM at 125 KIAS

Table 3: MDN G1000 Configuration Performance Data
(Note: Data represents Standard Atmospheric Conditions at maximum gross weight)

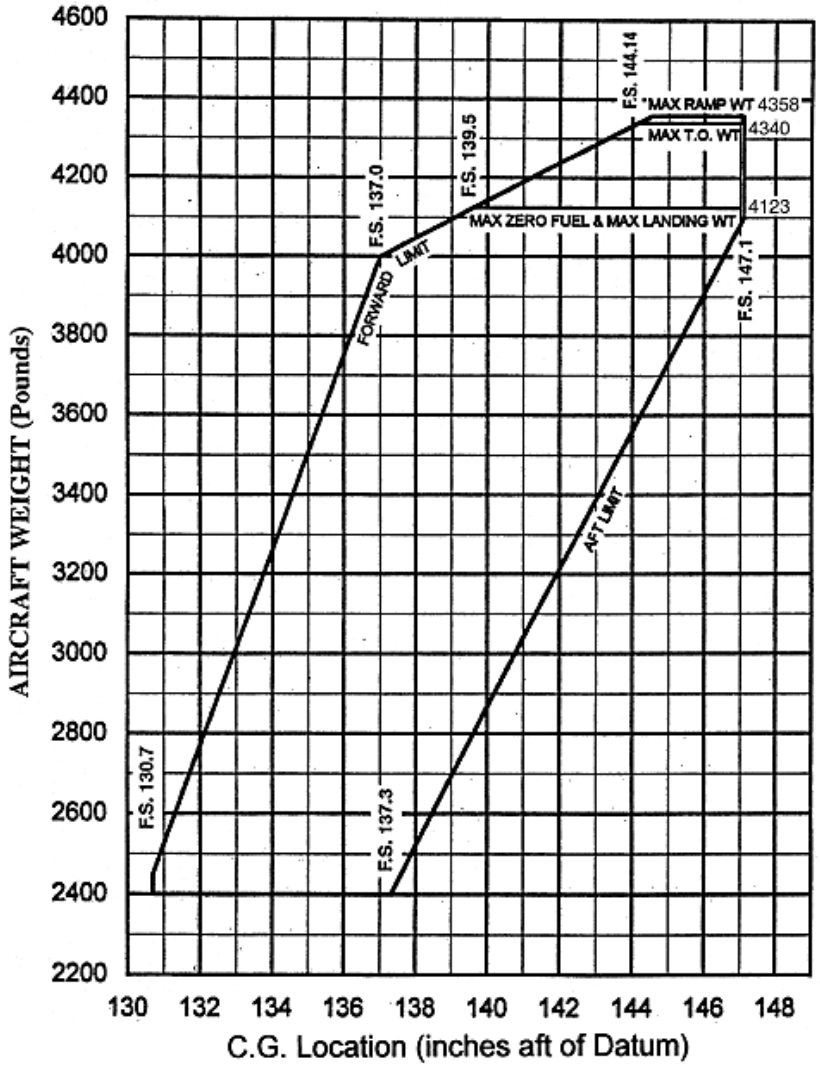
ADDITIONAL CONFIGURATIONS INCLUDED IN THIS VERSION (1.1):

1. Configuration: MRG (PA46-350-E1)

a. **Cockpit**



b. **Weight & Balance Data**



C.G. RANGE AND WEIGHT GRAPH
Figure 6-15

c. **Performance Data**

Altitude	MTOW	Maneuvering Speed (V_A)	Never Exceed Speed (V_{NE})	Max Structural Cruise Speed (V_{NO})	Max Cruise Speed (V_C)	Stall Speed (V_S)	Max Power Rate of Climb
Sea Level	4,340 lbs.	133 KIAS	198 KIAS	168 KIAS	165 KTAS at 2500 RPM	58 KIAS with Flaps-Down	1340 FPM at 110 KIAS
5,000 feet	4,340 lbs.	133 KIAS	198 KIAS	168 KIAS	170 KTAS at 2500 RPM	58 KIAS with Flaps-Down	1245 FPM at 110 KIAS

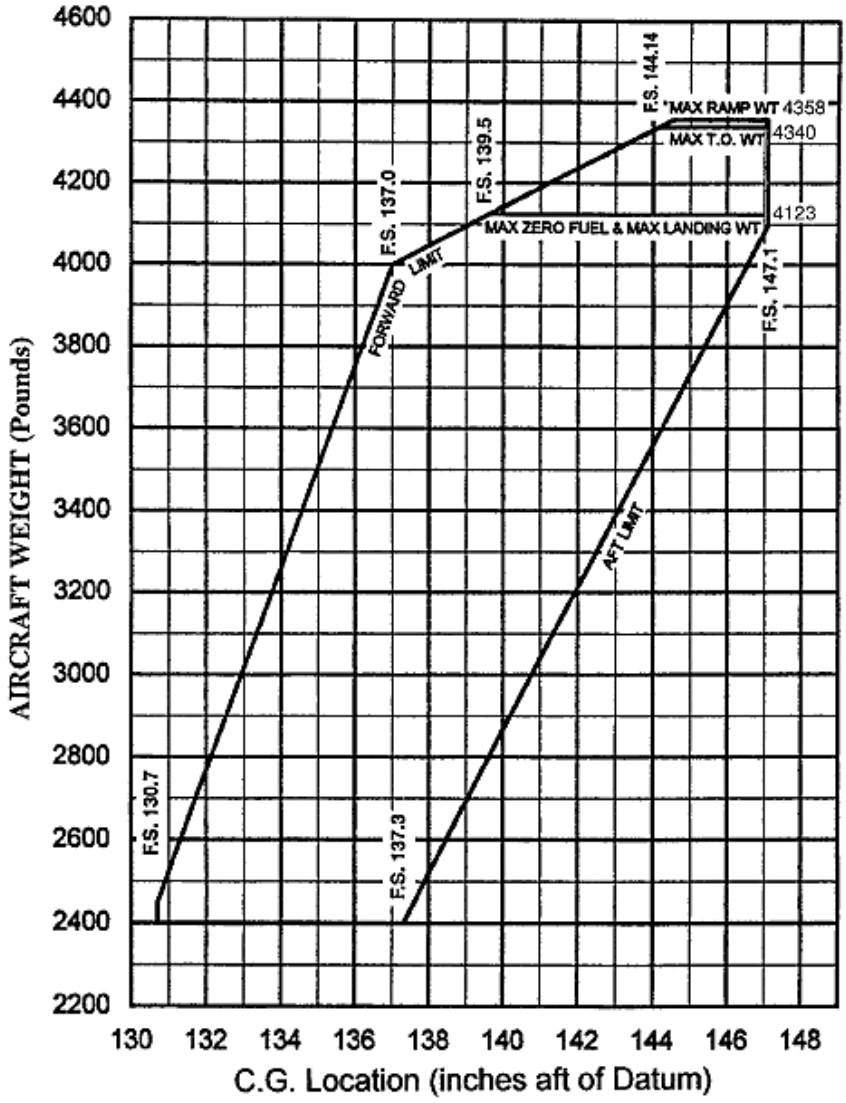
Table 4: MRG Avidyne Configuration Performance Data
(Note: Data represents Standard Atmospheric Conditions at maximum gross weight)

2. Configuration: MRG (PA46-350-G1)

a. **Cockpit**



b. **Weight & Balance Data**



C.G. RANGE AND WEIGHT GRAPH
Figure 6-15

c. **Performance Data**

Altitude	MTOW	Maneuvering Speed (V_A)	Never Exceed Speed (V_{NE})	Max Structural Cruise Speed (V_{NO})	Max Cruise Speed (V_C)	Stall Speed (V_S)	Max Power Rate of Climb
Sea Level	4,340 lbs.	133 KIAS	198 KIAS	168 KIAS	165 KTAS at 2500 RPM	58 KIAS with Flaps-Down	1340 FPM at 110 KIAS
5,000 feet	4,340 lbs.	133 KIAS	198 KIAS	168 KIAS	170 KTAS at 2500 RPM	58 KIAS with Flaps-Down	1245 FPM at 110 KIAS

Table 5: MRG G1000 Configuration Performance Data
(Note: Data represents Standard Atmospheric Conditions at maximum gross weight)

SECTION 6: VISUAL SYSTEM WITH VFR, IFR, DAY AND NIGHT CAPABILITIES

Mindstar Aviation Redbird G430/G1000 (MDN/MRG) and MDN/MRG Gauge "This is to certify that Mindstar Aviation, the owner and developer of the Redbird G430/G1000 (MDN/MRG) and MDN/MRG Gauge, and Redbird Flight Simulations, the hardware manufacture and component integrator, have demonstrated that the Mindstar Aviation software package is fully compatible with the Redbird Flight Simulations MDN and all configurations of that model. Mindstar Aviation and Redbird Flight Simulations can assure that the communications/transport data delay is not greater than 300 milliseconds and all analog and digital input signals meet the performance criteria established for the software performance by Redbird Flight Simulations."

Redbird MDN/MRG Visual System

The MDN/MRG visual system is capable of providing a field-of-view of a minimum of 45 degrees horizontally and 30 degrees vertically, simultaneously for each pilot, including adjustable cloud base and visibility in night, dusk and day scenes.

- The Redbird MDN/MRG provides a means of recording the visual response time for the visual system that is installed.
- The Redbird MDN/MRG visual system is free of optical discontinuities and artifacts that create non-realistic cues.
- The visual system is directly displayed on six (6) LCD monitors inside the cockpit enclosure, situated in an arc around the Pilot. Each monitor is 28 cm tall, and 47 cm wide. Based upon the designated Pilot Eye Point, these monitors provide a horizontal FOV of 220 degrees and a vertical FOV of 30 degrees.

Daylight: The visual system provides full color presentations and sufficient surfaces with appropriate textural cues to conduct a visual approach, landing and airport movement. Surface shading effects are consistent with the simulated sun position.

Twilight: The visual system provides full color presentations of reduced ambient intensity, sufficient surfaces with appropriate textural cues that include self-illuminated objects such as road networks, ramp lighting and airport signage, to conduct a visual approach, landing and airport movement. Scenes include a definable horizon and typical terrain characteristics such as fields, roads and bodies of water and surfaces illuminated by representative ownship lighting.

Night: The visual system provides the same as above except the portrayal of reduced ambient intensity, therefore, there is no ground cues that are not self-illuminating or illuminated by ownship lights.

Designated Eye Point: The designated Pilot Eye Point is located 52 cm from the center of the forward most external view monitor, 61 cm from the left most external view monitor and 24 cm from the ceiling of the simulator enclosure. This point is

roughly centered over the pilot's seat when it is adjusted to the forward most position, at a height consistent with the height of the pilot's head.

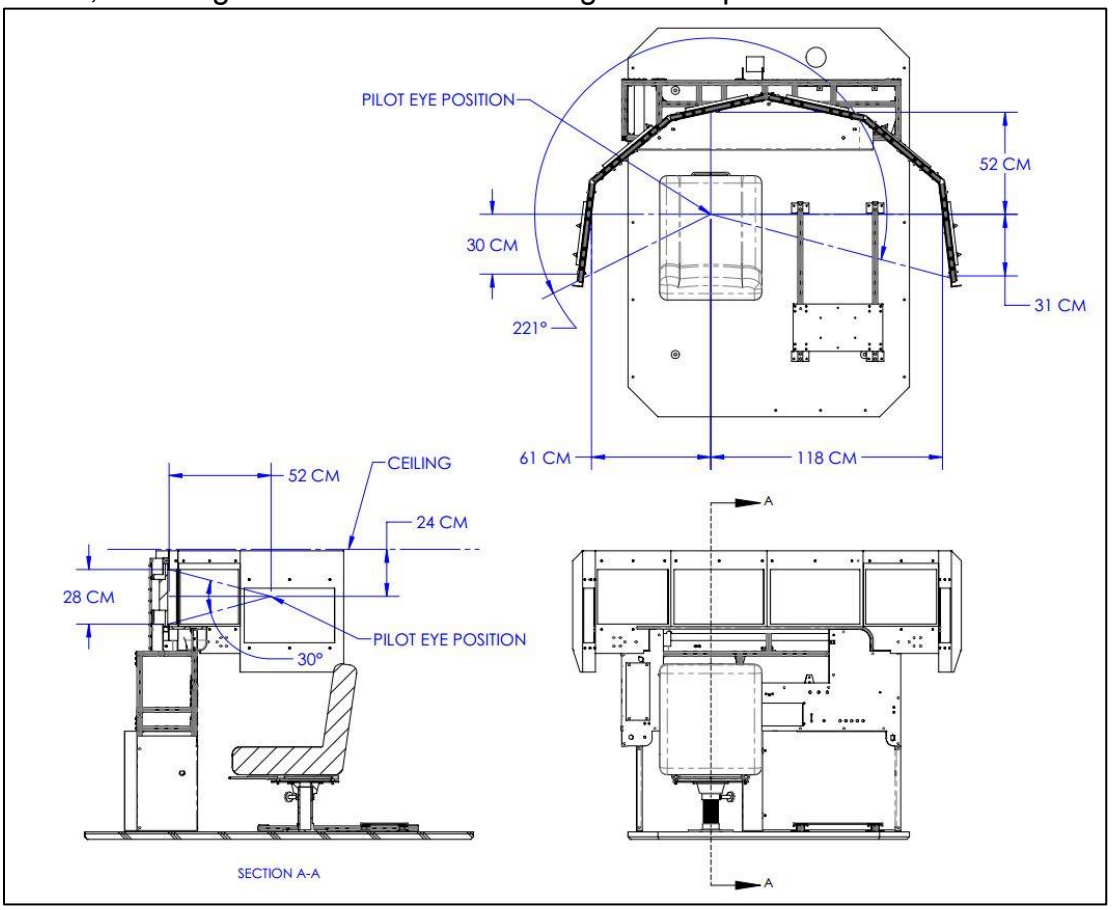


Figure 1: Designated Eye Point Diagram

SECTION 7: ATD FUNCTIONS AND MANEUVERS CHECKLIST

ATD FUNCTIONS AND MANEUVERS CHECKLIST

AVIATION TRAINING DEVICE ACTION CHECKLIST

Functions and Maneuvers	Yes, No, N/A
a. Preparation for Flight	
(1) Flight Deck Preflight	Yes
b. Pre-Takeoff	
(1) Engine start	Yes
(2) Brake operation	Yes
(3) Taxi operations and markings	Yes
c. Takeoff	
(1) **AIRPLANE Takeoff**	
(i) Power plant run-up and checks	Yes
(ii) Acceleration characteristics	Yes
(iii) Nose wheel and rudder steering	Yes
(iv) Effect of headwind/crosswind	Yes
(v) Instrument check	Yes
(vi) Landing gear, wing flap operation	Yes
(2) **HELICOPTER only Takeoff**	
(i) Power plant checks	N/A
(ii) From Hover	N/A
(iii) From Ground	N/A
(iv) Vertical	N/A
(v) Running	N/A
d. In-Flight Operation	
(1) **AIRPLANE In-Flight Operation**	
Runway and Takeoff	
(a) Normal and Maximum Performance Takeoffs	Yes
(b) Short and Soft Field Takeoff and Departures	Yes
(c) Normal and Maximum Performance Climbs	Yes
(d) Engine Failure and Emergency Procedures	Yes
Cruise	
(a) Performance Characteristics (speed vs. power)	Yes
(b) Normal, Climbing, and Descending Turns	Yes
(c) Performance Steep Turns	Yes

(d) Approach to Stalls, i.e. Stall Warning, Cruise, Takeoff & Approach and Landing Configurations	Yes
(e) In Flight Engine Shutdown (Multi-Engine only)	N/A
(f) In Flight Engine Start (Multi-Engine only)	N/A
Approach and Landing	
(a) Normal and Partial/No Flap Approach and Landings	Yes
(b) Short and Soft Field Approach and Landings	Yes
(c) Single Engine Approach and Landing (Multi-Engine)	N/A
Functions and Maneuvers	**Yes, No, N/A**
(2) **HELICOPTER In Flight Operations**	
(i) Hovering	N/A
(a) Forward	N/A
(b) Rearward	N/A
(c) Sideward	N/A
(d) Turns	N/A
(ii) Climb	N/A
(iii) Cruise	N/A
(a) Performance Characteristics	N/A
(b) Turns	N/A
(i) Recovery	N/A
(ii) Skidding	N/A
(iii) Slipping	N/A
(c) In Flight Engine Shutdown (Multi-Engine)	N/A
(d) In Flight Engine Start	N/A
(iv) Descent	N/A
e. Instrument Approaches	
(1) **Non-Precision**	
(i) GPS- WAAS (optional)	Yes
(ii) GPS- LAAS (optional)	Yes
(iii) All Engines Operating	Yes
(iv) One or More Engines Inoperative (as applicable)	N/A
(v) Approach Procedures (NDB, VOR, DME Arc, LOC/BC, LOC, LDA, SDF, ASR, LNAV/VNAV, GPS and LPV)	Yes
(2) **Precision**	
(i) ILS	Yes
(ii) Effects of Crosswind	Yes
(iii) With engine(s) inoperative	Yes
(iv) Missed Approach	Yes
(A) Normal	Yes
(B) With Engine(s) inoperative (as applicable)	N/A
(C) From Steep Glide Slope	Yes
f. Surface Operations (AIRPLANE-Post Landing)	

(1) Landing roll	Yes
(2) Braking Operation	Yes
(3) Reverse thrust Operation, if applicable	Yes
g. Any Flight Phase	
(1) **Aircraft and Power Plant Systems Functions and Simulated Failures**	
(i) Electrical	Yes
(ii) Flaps (Airplane)	Yes
(iii) Fuel & Oil	Yes
(iv) Landing Gear	Yes
(2) **Flight Management and Guidance Systems**	
(i) Two Axis Auto Pilot (AATD only)	Yes
Functions and Maneuvers	**Yes, No, or N/A**
(ii) Flight Director (AATD only) / System Displays	Yes
(iii) Navigation Systems and optional display configurations	Yes
(iv) Stall Warning Avoidance (Airplane)	Yes
(v) Multi-Function Displays PFD/MFD	Yes
(3) **Airborne Procedures**	
(i) Holding	Yes
(4) **Simulated Turbulence in Flight**	
(5) **Engine Shutdown and Parking**	
(i) Systems operation	Yes
(ii) Parking Brake Operation (Airplane)	Yes
h. Training device capable of replicating any emergency procedures provided in the Aircraft Flight Manual "checklists" for the model or aircraft configuration represented.	Yes

Any changes or modifications to this device which have not been reviewed, evaluated, and approved in writing by General Aviation and Commercial Division, AFS-800 may terminate FAA approval of this aviation training device.